Shi Qiu

First-year M.S. Student in Computer Science, NYU Courant
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Education

New York University, Courant Institute — New York, NY
M.S. in Computer Science — *Sep. 2026 – Expected May 2028*

Xi'an Jiaotong-Liverpool University — Suzhou, China
B.Eng. in Electrical Engineering, Upper Second-Class Honours — *Sep. 2022 – Jun. 2026*

Research Interests

Trustworthy LLMs and agentic systems: reliable evaluation, verifiable correction, and evidence-grounded decision-making. My work connects machine unlearning and behavioral control with auditable, human-supervised agents for scientific and engineering design.

Publications and Manuscripts

Published and Accepted

[1] **Shi Qiu**, Fanfan Lin, Xinze Li. *PE-GPT 2.0: A Trustworthy Multi-Agent Design Framework for Power Converters*. **IRAI 2026**, accepted; **IEEE IES SYPA Travel Award**; [3-min video] [code]

[2] Haichao Zhang, Chong Zhang, Wen Wang, **Shi Qiu**, Peiyu Hu, Jia Wang. *Explain, Then Forget: Recommendation Unlearning under Asymmetric Access Constraint*. **ICDM 2026**, accepted; [code]

[3] Haichao Zhang, Chong Zhang, Can Wang, **Shi Qiu**, Jia Wang. *Explaining Graph Recommendations via Counterfactual Support Sets*. **ICONIP 2026**, accepted; [code]

[4] Can Wang, **Shi Qiu**, Haichao Zhang. *Topology-Aware Inverse Preference Learning for Robust Hybrid Voting Systems*. **ICONIP 2026**, accepted; [code]

[5] Haichao Zhang, Chong Zhang, Peiyu Hu, **Shi Qiu**, Jia Wang. *Customized Retrieval-Augmented Generation with LLM for Debiasing Recommendation Unlearning*. **ICDM 2025**; [code]

Research Manuscripts

*Equal contribution

[6] Can Wang*, **Shi Qiu***, Fanfan Lin. *PowerFix: Physics-Guided Iterative Repair for LLM-Assisted Power Converter Design*. AAAI 2027 Innovative Applications of AI (IAAI), under review; [code]

[7] **Shi Qiu**, Fanfan Lin, Xinze Li, Kezhi Mao, Junhua Liu. *PE-Bench: A Benchmark for Evaluating Agent-Based Power Electronics Design Systems*. Unpublished manuscript, 2026; [code]

[8] Haichao Zhang, **Shi Qiu**, Hao Chen, Dongxiao Hu, Chong Zhang, Elliott Wen, Jia Wang. *Distilled Prompt Tuning for Efficient GNN-based Recommendation Unlearning*. ACM Transactions on Information Systems (TOIS), under review; [code]

Research Experience

Zhejiang University, ZJU-UIUC Institute — Haining, China
Research Assistant | Supervisor: Prof. Fanfan Lin — *Feb. 2026 – Present*

- **Reliable evaluation (PE-Bench).** Co-developed 78 executable tasks across five converter families, checking specification compliance, physical consistency, component feasibility, and supporting evidence. For a generic multi-agent baseline, simulation-only checks reported 52.4% success versus 23.5% under the full evaluator.
- **Verifiable repair (PowerFix).** Co-developed physics-guided diagnosis, constrained patch compilation, and isolated validation for LLM-generated flyback designs. In pre-pilot evaluation on 96 failed designs, achieved 78.7% verified repair success versus 30.1% for the strongest baseline, averaged equally across three failure groups.
- **Auditable agents (PE-GPT 2.0).** Built a multi-agent design workflow integrating equation checks, component verification, PLECS simulation, and review gates; linked generated designs to engineering evidence, correction histories, and final review decisions.

Research Experience (Continued)

Recommendation Algorithm Research Group China
Research Member, Recommendation Unlearning *Apr. 2025 – Present*

- **Controllable model behavior.** Contributed to recommendation unlearning through RAG-based debiasing, dual-teacher prompt distillation, and output-level interventions under asymmetric access, studying selective forgetting while preserving recommendation utility.
- **User-centered evaluation (AgentURE).** Leading a project to assess whether forgetting changes the user's recommendation experience. Designed measures of perceived forgetting, target exposure, semantic residue, and retained utility, with a protocol covering two datasets and 1,000 forget requests per dataset.
- **Exploratory prototypes.** Investigated dual-process memory and hierarchical preference alignment in DURE; implemented prompt-unlearning and codebook-based generative recommendation experiments.

Xi'an Jiaotong-Liverpool University Suzhou, China
iGEM Software and Modeling | Supervisor: Prof. Guoxia Han *Jan. 2023 – Present*

- **Evidence-grounded design (AMPilot, iGEM 2026).** Developed an LLM/RAG workbench for antimicrobial peptide design with natural-language task routing and retrieval over AMP, literature, and formulation evidence; surfaced sources, design constraints, and uncertainty for candidate review.
- **Human oversight and traceability.** Built a FastAPI backend and event-driven interface with versioned run records and checkpointed execution, enabling resumable tasks, audit trails, human approvals, and structured experimental feedback.
- **Controlled outputs.** Integrated candidate triage and review of formulation, production, and construct options; implemented export approval and biosafety policies that distinguish model predictions from experimental evidence. Public prototype; current development maintained privately.

Industry Experience

Microsoft China
Artificial Intelligence Engineer Intern *Jun. 2025 – Nov. 2025*

- Designed, fine-tuned, and deployed LLM components for real-time public-opinion monitoring, including continuous-learning pipelines and sentiment analysis over 40,000+ news articles.
- Improved model pipelines and processing latency for multimodal video and image editing workflows.

Honors and Competitions

- **IEEE IES SYPA Travel Award** 2026
  One of 10 award recipients; PE-GPT 2.0. [3-min research video]
- **China Scholarship Council (CSC) Scholarship** **2026**
  Awarded 48-month doctoral funding for PhD study at McGill University; commencement deferred due to study-permit delays.
- **iGEM Gold Medal**, CLARITEIN; software and modeling lead. 2023
- **iGEM Bronze Medal**, Eco MEGA; computational pipeline contributor. 2024
- **Kaggle, Predicting F1 Pit Stops** (Playground S6E5): 544/3,023, top 18%. 2026
- **Mathematical Contest in Modeling**: Successful Participant (S); team leader. 2025, 2026

Leadership and Service

President, iOS Club, Xi'an Jiaotong-Liverpool University Sep. 2022 – Jun. 2024
Organized 5+ technical events per semester; developed collaborations with Apple Suzhou and Duke Kunshan University, increasing active membership by 30%.

Technical Skills

Programming: Python, C/C++, SQL, Bash, JavaScript.
ML: PyTorch, Transformers, scikit-learn; RAG, prompt tuning, LoRA, knowledge distillation.
Systems and Evaluation: FastAPI, Git, Docker, Linux/HPC, PLECS.